

Solutions for GATE 2025 Biomedical Engineering (BM)

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I. SOLUTION FOR Q62

Q.62	If the gyromagnetic ratio of proton is 42.58 MHz per Tesla, the resonance frequency for protons in a magnetic field of 1.5 Tesla is _____ MHz. (rounded off to two decimal places)
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Given:

$$\gamma = 42.58 \text{ MHz/T}, \quad B = 1.5 \text{ T}$$

The resonance (Larmor) frequency is given by

$$f = \gamma B$$

Substituting the values,

$$f = (42.58)(1.5)$$

$$f = 63.87 \text{ MHz}$$

$$\boxed{63.87 \text{ MHz}}$$

$$\sigma_0 = 2 + 2 \log_e(4)$$

Since,

$$\log_e(4) = 1.3863$$

$$\sigma_0 = 2 + 2(1.3863)$$

$$\sigma_0 = 4.7726$$

Now, at $t = 8$ weeks,

$$\sigma = 4.7726 - 2 \log_e(8)$$

Since,

$$\log_e(8) = 2.0794$$

$$\sigma = 4.7726 - 2(2.0794)$$

$$\sigma = 4.7726 - 4.1588$$

$$\sigma = 0.6138 \text{ MPa}$$

Therefore,

$$\boxed{\sigma = 0.61 \text{ MPa}}$$

II. SOLUTION FOR Q61

Q.61	Strength (σ) of an implanted suture is given by the equation $\sigma = \sigma_0 - 2 \log_e(t) \text{ for } t \geq 1,$ where σ_0 represents the original strength and time t is measured in weeks. If the strength of the suture is 2 MPa at 4 weeks, what will be its strength (in MPa) at 8 weeks? _____ (rounded off to two decimal places)
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Given:

$$\sigma = \sigma_0 - 2 \log_e(t)$$

At $t = 4$ weeks, the strength is

$$\sigma = 2 \text{ MPa}$$

Substituting the values,

$$2 = \sigma_0 - 2 \log_e(4)$$

III. SOLUTION FOR Q58

Q.58	For pulsatile blood flow through an artery of internal diameter 20 mm, wall thickness 1 mm and Young's Modulus 1 MPa, what is the wave speed in metres per second? _____ (rounded off to one decimal place). Assume the density of blood to be 1050 kg/m ³ .
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Given:

$$E = 1 \text{ MPa} = 1 \times 10^6 \text{ Pa}$$

$$h = 1 \text{ mm} = 0.001 \text{ m}$$

$$D = 20 \text{ mm} = 0.02 \text{ m}$$

$$\rho = 1050 \text{ kg/m}^3$$

Using the Moens–Korteweg equation,

$$c = \sqrt{\frac{Eh}{\rho D}}$$

Substituting the values,

$$c = \sqrt{\frac{(1 \times 10^6)(0.001)}{(1050)(0.02)}}$$

$$c = \sqrt{\frac{1000}{21}}$$

$$c = \sqrt{47.619}$$

$$c = 6.9007 \text{ m/s}$$

Therefore,

$c = 6.9 \text{ m/s}$

IV. SOLUTION FOR Q57

Q.57	Newton Raphson method is used to solve the equation $x^3 - 2x - 5 = 0$ with an initial guess $x_0 = 3$. What is x_1 , the value after ONE iteration? _____ (rounded off to two decimal places)
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Given:

$$f(x) = x^3 - 2x - 5$$

Newton-Raphson formula:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

The derivative is:

$$f'(x) = 3x^2 - 2$$

For the initial guess,

$$x_0 = 3$$

Calculate $f(3)$:

$$f(3) = 3^3 - 2(3) - 5$$

$$f(3) = 27 - 6 - 5$$

$$f(3) = 16$$

Calculate $f'(3)$:

$$f'(3) = 3(3^2) - 2$$

$$f'(3) = 27 - 2$$

$$f'(3) = 25$$

Substituting into the Newton-Raphson formula:

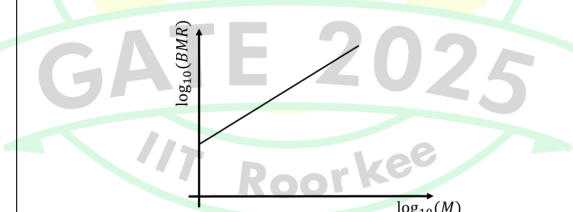
$$x_1 = 3 - \frac{16}{25}$$

$$x_1 = 3 - 0.64$$

$$x_1 = 2.36$$

Therefore,

$x_1 = 2.36$

Q.56	The plot of $\log_{10}(BMR)$ as a function of $\log_{10}(M)$ is a straight line with slope 0.75, where M is the mass of the person and BMR is the Basal Metabolic Rate. If a child with $M = 10$ kg has a $BMR = 600$ kcal/day, the BMR for an adult with $M = 100$ kg is _____ kcal/day. (rounded off to the nearest integer)
	

V. SOLUTION FOR Q56

Given:

$$\log_{10}(BMR) = 0.75 \log_{10}(M) + C$$

Therefore,

$$BMR \propto M^{0.75}$$

Using the ratio form:

$$\frac{BMR_2}{BMR_1} = \left(\frac{M_2}{M_1} \right)^{0.75}$$

Given:

$$M_1 = 10 \text{ kg}$$

$$BMR_1 = 600 \text{ kcal/day}$$

$$M_2 = 100 \text{ kg}$$

Substituting,

$$\frac{BMR_2}{600} = \left(\frac{100}{10} \right)^{0.75}$$

$$\frac{BMR_2}{600} = 10^{0.75}$$

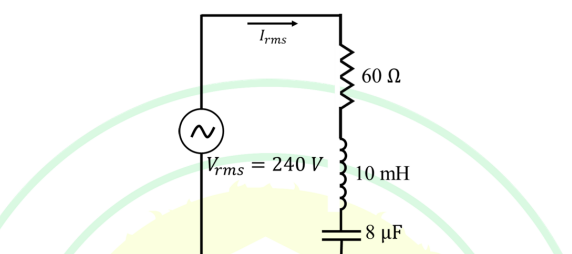
$$\frac{BMR_2}{600} = 5.6234$$

$$BMR_2 = 600 \times 5.6234$$

$$BMR_2 = 3374.04$$

Therefore,

$$\boxed{3374 \text{ kcal/day}}$$

Q.55	For the RLC circuit shown below, the root mean square current (I_{rms}) at the resonance frequency is _____ amperes. (rounded off to the nearest integer)
	

VI. SOLUTION FOR Q55

Given:

$$V_{rms} = 240 \text{ V}$$

$$R = 60 \Omega$$

At resonance in a series RLC circuit,

$$X_L = X_C$$

Hence, the impedance is purely resistive:

$$Z = R$$

Using Ohm's law,

$$I_{rms} = \frac{V_{rms}}{R}$$

Substituting the values,

$$I_{rms} = \frac{240}{60}$$

$$I_{rms} = 4 \text{ A}$$

Therefore,

$$\boxed{I_{rms} = 4 \text{ A}}$$

VII. SOLUTION FOR Q54

Given:

$$R = 1 \text{ k}\Omega = 1000 \Omega$$

$$C = 0.1 \mu\text{F} = 10^{-7} \text{ F}$$

This is actually the classic op-amp RC phase-shift / relaxation oscillator whose frequency is often approximated by

$$f = \frac{1}{2\pi RC}$$

Substituting the values,

$$f = \frac{1}{2\pi(1000)(10^{-7})}$$

$$f = \frac{1}{6.283 \times 10^{-4}}$$

Q.54	The frequency of the oscillator circuit shown in the figure below is _____ kHz. (rounded off to two decimal places) Given: $R = 1 \text{ k}\Omega$; $R_1 = 2 \text{ k}\Omega$; $R_2 = 6 \text{ k}\Omega$; $C = 0.1 \mu\text{F}$

$$f = 1591.55 \text{ Hz}$$

$$f = 1.59 \text{ kHz}$$

Therefore,

$$\boxed{1.59 \text{ kHz}}$$

VIII. SOLUTION FOR Q53

Given:

$$R(T) = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$$

$$\beta = 3100 \text{ K}$$

$$T = 310 \text{ K}$$

Taking logarithm,

Q.53	Resistance R of a thermistor varies as a function of temperature T such that $R(T) = R_0 \exp \left[\beta \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$ where $\beta = 3100 \text{ K}$ and R_0 and T_0 are positive constants. If the relative error in measuring R is 10%, what is the relative error (in percentage) in measuring 310 K temperature? _____ (rounded off to the nearest integer)
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$$\ln R = \ln R_0 + \beta \left(\frac{1}{T} - \frac{1}{T_0} \right)$$

Differentiating,

$$\frac{dR}{R} = -\frac{\beta}{T^2} dT$$

Taking magnitudes,

$$\left| \frac{dR}{R} \right| = \frac{\beta}{T^2} |dT|$$

Therefore,

$$\frac{|dT|}{T} = \frac{T}{\beta} \left| \frac{dR}{R} \right|$$

Given,

$$\frac{|dR|}{R} = 10\% = 0.10$$

Substituting,

$$\frac{|dT|}{T} = \frac{310}{3100} (0.10) = 0.01$$

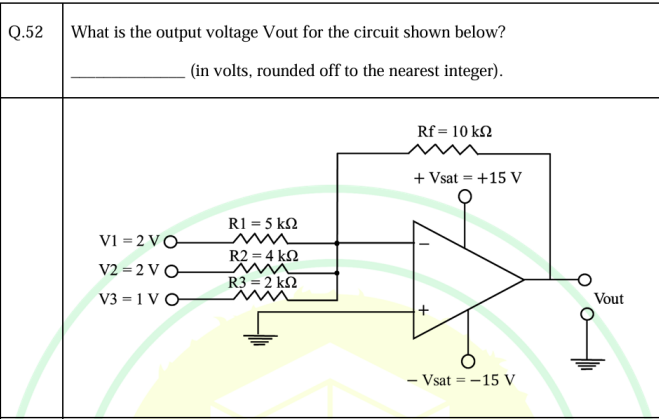
Converting to percentage,

$$\text{Relative error in } T = 0.01 \times 100$$

$$= 1\%$$

Therefore,

$$\boxed{1\%}$$



IX. SOLUTION FOR Q52

Given:

$$R_f = 10 \text{ k}\Omega$$

$$V_1 = 2 \text{ V}, \quad R_1 = 5 \text{ k}\Omega$$

$$V_2 = 2 \text{ V}, \quad R_2 = 4 \text{ k}\Omega$$

$$V_3 = 1 \text{ V}, \quad R_3 = 2 \text{ k}\Omega$$

For an inverting summing amplifier,

$$V_{out} = -R_f \left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right)$$

Substituting the values,

$$V_{out} = -10 \text{ k}\Omega \left(\frac{2}{5 \text{ k}\Omega} + \frac{2}{4 \text{ k}\Omega} + \frac{1}{2 \text{ k}\Omega} \right)$$

$$V_{out} = -10 (0.4 + 0.5 + 0.5)$$

$$V_{out} = -14 \text{ V}$$

Since

$$-15 \text{ V} < -14 \text{ V} < +15 \text{ V}$$

the output does not saturate.

Therefore,

$$V_{out} = -14 \text{ V}$$

Q.51 For an 8-bit Digital to Analog Converter (DAC), the binary input 0000 0000 results in 0 V, and 1111 1111 results in 5 V. The output of the DAC for an input of 1011 0111 is _____ (in volts, rounded off to two decimal places).

X. SOLUTION FOR Q51

Given:

$$0000 \ 0000 \rightarrow 0 \text{ V}$$

$$1111 \ 1111 \rightarrow 5 \text{ V}$$

Input code:

$$1011 \ 0111$$

Converting to decimal,

$$10110111_2 = 128 + 32 + 16 + 4 + 2 + 1$$

$$= 183$$

For an 8-bit DAC,

$$V_{out} = \frac{\text{Digital Code}}{255} \times 5$$

$$V_{out} = \frac{183}{255} \times 5$$

$$V_{out} = 3.5882 \text{ V}$$

$$V_{out} \approx 3.59 \text{ V}$$

Therefore,

$$\boxed{3.59 \text{ V}}$$

XI. SOLUTION FOR Q50

Given:

Biomaterial	G' (Pa)	G'' (Pa)
P	1000	50
Q	1000	80
R	1000	100
S	1000	0

The storage modulus G' represents the elastic (solid-like) behavior, while the loss modulus G'' represents the viscous (fluid-like) behavior.

A material behaves as a viscoelastic solid when

Q.50	At a given frequency, the storage modulus (G') and loss modulus (G'') of four biomaterials are shown in the table below. Which of the following option(s) is/are CORRECT?			
	Biomaterial	G' (Pa)	G'' (Pa)	
	P	1000	50	
	Q	1000	80	
	R	1000	100	
	S	1000	0	
(A)	P, Q and R are viscoelastic solids			
(B)	P, Q and R are viscoelastic fluids			
(C)	R and S are viscoelastic solids			
(D)	S is an elastic solid			

$$G' > G''$$

and as a viscoelastic fluid when

$$G'' > G'.$$

For biomaterial P,

$$1000 > 50$$

Therefore, P is a viscoelastic solid.

For biomaterial Q,

$$1000 > 80$$

Therefore, Q is a viscoelastic solid.

For biomaterial R,

$$1000 > 100$$

Therefore, R is a viscoelastic solid.

For biomaterial S,

$$G'' = 0$$

Hence, there is no viscous component and S behaves as a purely elastic solid.

Checking the options:

- (A) P, Q and R are viscoelastic solids
 (B) P, Q and R are viscoelastic fluids
 (C) R and S are viscoelastic solids
 (D) S is an elastic solid

Correct
Incorrect
Incorrect
Correct

Therefore,

Q.49	A Newtonian fluid of viscosity μ is undergoing laminar flow in a tube of radius R . For a constant pressure drop per unit length along the tube, which of the following statement(s) is/are CORRECT? Note: r represents the radial position; assume no slip condition at the boundaries.	
(A)	The radial profile of the flow velocity is proportional to $(1 - \frac{r^2}{R^2})$	
(B)	The radial profile of the flow velocity is proportional to $(1 - \frac{r^3}{R^3})$	
(C)	Shear stress is zero at $r = R$	
(D)	Shear stress is zero at $r = 0$	

(A) and (D)

XII. SOLUTION FOR Q49

For laminar flow of a Newtonian fluid in a circular tube, the Hagen–Poiseuille velocity profile is

$$u(r) = \frac{\Delta P}{4\mu L}(R^2 - r^2)$$

or

$$u(r) \propto \left(1 - \frac{r^2}{R^2}\right).$$

Therefore,

$$u(r) \propto \left(1 - \frac{r^2}{R^2}\right)$$

Hence, option (A) is **correct**.

The profile

$$u(r) \propto \left(1 - \frac{r^3}{R^3}\right)$$

is not valid for Hagen–Poiseuille flow. Therefore, option (B) is **incorrect**.

For a Newtonian fluid,

$$\tau = \mu \frac{du}{dr}$$

Differentiating the velocity profile,

$$\frac{du}{dr} = -\frac{\Delta P}{2\mu L}r$$

Hence,

$$\tau = -\frac{\Delta P}{2L}r$$

Thus,

$$\tau \propto r.$$

At the centerline,

$$r = 0$$

$$\tau = 0$$

Therefore, option (D) is **correct**.

At the wall,

$$r = R$$

$$\tau = -\frac{\Delta P}{2L}R \neq 0$$

Therefore, option (C) is **incorrect**.

Hence,

(A) and (D)

XIII. SOLUTION FOR Q47

Q.47	A discrete linear time invariant system has an impulse response function given by $h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{3}\delta[n-2]$. For input signal $x[n] = \delta[n] + \delta[n-1]$, which of the following option(s) is/are CORRECT for the output signal $y[n]$?
(A)	$y[2] = 11/6$
(B)	$y[2] = 5/6$
(C)	$y[k] = 0$ for all $k \geq 3$
(D)	$y[k] = 0$ for all $k \geq 4$

Given:

$$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{3}\delta[n-2]$$

$$x[n] = \delta[n] + \delta[n-1]$$

For an LTI system,

$$y[n] = x[n] * h[n]$$

Using the convolution property,

$$(\delta[n] + \delta[n-1]) * h[n] = h[n] + h[n-1]$$

Now,

$$h[n-1] = \delta[n-1] + \frac{1}{2}\delta[n-2] + \frac{1}{3}\delta[n-3]$$

Therefore,

$$y[n] = \delta[n] + \left(1 + \frac{1}{2}\right)\delta[n-1] + \left(\frac{1}{2} + \frac{1}{3}\right)\delta[n-2] + \frac{1}{3}\delta[n-3]$$

$$y[n] = \delta[n] + \frac{3}{2}\delta[n-1] + \frac{5}{6}\delta[n-2] + \frac{1}{3}\delta[n-3]$$

Hence,

$$y[2] = \frac{5}{6}$$

and

$$y[3] = \frac{1}{3}$$

$$y[k] = 0, \quad k \geq 4$$

Therefore,

(B) and (D)

XIV. SOLUTION FOR Q45

Q.45	Which of the following statement(s) about pulse oximetry is/are TRUE?
(A)	Oxygenated hemoglobin absorbs less infrared light (900 nm) compared to deoxygenated hemoglobin
(B)	Oxygenated hemoglobin absorbs less visible light (600 nm) compared to deoxygenated hemoglobin
(C)	Oxygenated hemoglobin absorbs more visible light (600 nm) compared to deoxygenated hemoglobin
(D)	Oxygenated hemoglobin absorbs more infrared light (900 nm) compared to deoxygenated hemoglobin

Solution:

Principle of Pulse Oximetry:

Pulse oximetry uses two wavelengths of light:

- Red light (≈ 660 nm)
- Infrared light (≈ 940 nm)

The absorption characteristics are:

- Oxygenated hemoglobin (HbO₂) absorbs less red/visible light than deoxygenated hemoglobin (Hb).
- Oxygenated hemoglobin (HbO₂) absorbs more infrared light than deoxygenated hemoglobin (Hb).

Therefore,

- (A) False
- (B) True
- (C) False
- (D) True

Hence,

(B) and (D)

XV. SOLUTION FOR Q44

Q.44	Two systems U and V are defined as shown below. Which of the following statement(s) is/are CORRECT?
	$\begin{array}{ccc} x(t) & \xrightarrow{\text{System } U} & y(t) = x(t)^2 + 1 \\ x(t) & \xrightarrow{\text{System } V} & y(t) = x(t + 1) \end{array}$
(A)	System U is nonlinear; System V is linear
(B)	System U is causal; System V is noncausal
(C)	System U is noncausal; System V is causal
(D)	System U is nonlinear; System V is nonlinear

Solution:

For System U ,

$$y(t) = x^2(t) + 1$$

- **Linearity:** Since the output contains a squared term and a constant offset,

$$T\{ax_1(t) + bx_2(t)\} = (ax_1(t) + bx_2(t))^2 + 1 \neq aT\{x_1(t)\} + bT\{x_2(t)\},$$

therefore System U is **nonlinear**.

- **Causality:** The output at time t depends only on the present input $x(t)$. Hence, System U is **causal**.

For System V ,

$$y(t) = x(t + 1)$$

- **Linearity:**

$$T\{ax_1(t) + bx_2(t)\} = (ax_1(t) + bx_2(t))(t+1) = ax_1(t+1) + bx_2(t+1) = aT\{x_1(t)\} + bT\{x_2(t)\}.$$

Therefore, System V is **linear**.

- **Causality:** The output at time t depends on the future input value $x(t + 1)$. Hence, System V is **noncausal**.

Therefore,

System U is nonlinear and causal

System V is linear and noncausal

Hence, the correct options are

(A) and (B)

XVI. SOLUTION FOR Q42

Q.42	Which of the following is/are CORRECT about the value of $y = \log_e(-e^x)$, where x is a real number.
(A)	y is real valued
(B)	y is complex valued
(C)	$y = x \pm in\pi$, where n is an odd integer and $i = \sqrt{-1}$
(D)	$y = x \pm in\pi$, where n is an even integer and $i = \sqrt{-1}$

Solution:

Given

$$y = \log_e(-e^x) = \ln(-e^x).$$

Since $e^x > 0$ for all real x ,

$$-e^x < 0.$$

Therefore, y is not real-valued and must be evaluated using the complex logarithm:

$$\ln(-e^x) = \ln(e^x e^{i(2k+1)\pi}) = x + i(2k+1)\pi, \quad k \in \mathbb{Z}.$$

Let $n = 2k + 1$, where n is an odd integer. Then

$$y = x \pm in\pi, \quad n \text{ is odd.}$$

Hence,

- y is complex valued.
- $y = x \pm in\pi$, where n is an odd integer.

Therefore, the correct options are

(B) and (C).

Q.41	Gelatin solutions P and Q of concentrations 3.5% and 0.5%, respectively, need to be mixed to obtain a 3% solution R. How much volume (in ml) of P needs to be mixed with 100 ml of Q to obtain R? (Choose the correct option) All concentrations are in units of weight/volume.
(A)	100
(B)	250
(C)	500
(D)	1000

XVII. SOLUTION FOR Q41

Solution:

Let x ml of solution P (3.5%) be mixed with 100 ml of solution Q (0.5%).

The amount of gelatin in solution P is

$$\frac{3.5}{100}x = 0.035x \text{ g.}$$

The amount of gelatin in solution Q is

$$\frac{0.5}{100} \times 100 = 0.5 \text{ g.}$$

Hence, the total amount of gelatin in the mixture is

$$0.035x + 0.5.$$

The total volume of the mixture is

$$x + 100.$$

Since the required concentration is 3%,

$$\frac{0.035x + 0.5}{x + 100} = 0.03.$$

Multiplying both sides by $(x + 100)$,

$$0.035x + 0.5 = 0.03x + 3.$$

$$0.005x = 2.5.$$

$$x = \frac{2.5}{0.005} = 500.$$

Therefore,

$$\boxed{x = 500 \text{ ml}}.$$

Hence, the correct option is

$$\boxed{\text{(C) } 500}.$$

Q.32	If $f(x) = x - \frac{1}{x}$, the value of $\lim_{\Delta x \rightarrow 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}$ is _____.
(A)	0
(B)	$\frac{1}{2}$
(C)	1
(D)	2

XVIII. SOLUTION FOR Q32

Solution:

Given

$$f(x) = x - \frac{1}{x}.$$

The given limit is

$$\lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x},$$

which is the definition of the derivative of $f(x)$ at $x = 1$. Therefore,

$$\lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x} = f'(1).$$

Differentiating $f(x)$,

$$f'(x) = \frac{d}{dx} \left(x - \frac{1}{x} \right) = 1 + \frac{1}{x^2}.$$

Evaluating at $x = 1$,

$$f'(1) = 1 + \frac{1}{1^2} = 2.$$

Hence,

$$\boxed{\lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x) - f(1)}{\Delta x} = 2.}$$

Therefore, the correct option is

$$\boxed{\text{(D) } 2}.$$